Evaluation and Recommendation Framework for Large Language Models Based on Reasoning and Answer Alignment

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Background

Large Language Models (LLMs) are widely used across multiple application domains

Common use cases include:

- Question Answering
- Programming Assistance
- Logical Reasoning
- Decision-Making Systems

Most existing evaluation techniques assess model performance based primarily on final answer correctness

Current benchmarks focus on output accuracy rather than evaluating the reasoning process behind the response.

Need for Reasoning-Aware Evaluation in LLMs

A model may produce the correct output despite flawed or inconsistent reasoning
Logically sound reasoning improves reliability in real-world deployments

A systematic evaluation framework is required to assess:

- Correctness of the final answer
- Quality and validity of the reasoning process

Many applications demand domain-specific and expert-level knowledge
Reasoning evaluation helps determine appropriate use of domain expertise

The Rashomon Effect as an Interpretability Problem

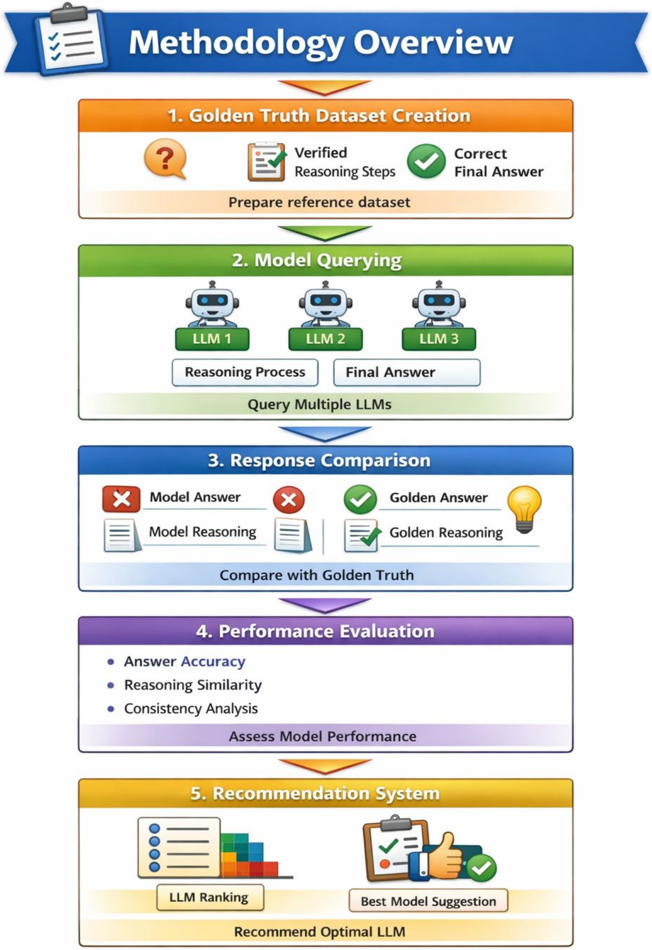
- The Rashomon Effect in ML: multiple models can achieve equivalent predictive accuracy while relying on entirely different internal logic
- In LLMs this manifests as: same answer, different reasoning chains — which model do you trust, and on what basis?
- This is not just a performance question, it is a fundamental interpretability question
- High answer accuracy across models masks divergence in reasoning quality — accuracy metrics alone cannot surface this
- Our framework exposes this hidden divergence by evaluating reasoning alignment independently of answer correctness

The Rashomon Effect is the theoretical backbone of this project: we are not asking which model is most accurate, we are asking which model reasons most like a human expert

Chain-of-Thought as a Window into Interpretability

- Chain-of-Thought (CoT) prompting instructs models to externalize their reasoning step by step before arriving at an answer
- This is currently the most accessible mechanism for probing LLM reasoning behavior.
- A model's explanation can be plausible and human-like while being entirely disconnected from what actually drove the output.
- Our project operates at the level of stated reasoning, and we treat this honestly as measuring reasoning surface alignment, not ground truth interpretability

Proposed METHODOLOGY



Golden Truth

JSON

```json
{
  "question_id": "",
  "task_domain": "",
  "difficulty_level": "",

  "question": "",

  "golden_reasoning_steps": [
    {
      "step_number": 1,
      "description": ""
    },
    {
      "step_number": 2,
      "description": ""
    },
    {
      "step_number": 3,
      "description": ""
    }
  ],

  "golden_final_answer": "",

  "expected_knowledge_domain": "",

  "reasoning_type": "",

  "evaluation_criteria": {
    "must_include_concepts": [],
    "common_failure_modes": [],
    "logical_dependencies": []
  }
}
```

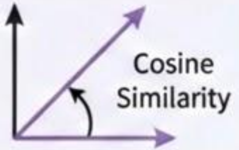
Reasoning Alignment Score (RAS)

Let:

- RR_i = Model-generated reasoning
- RR_g = Golden reasoning
- $E(R)E(R)$ = embedding of reasoning text

$$RAS_i = \frac{E(R_i) \cdot E(R_g)}{\|E(R_i)\| \|E(R_g)\|}$$

This is cosine similarity between reasoning embeddings.



$$0 \leq RAS_i \leq 1$$

Step-wise Logical Matching Score (SLMS)

Let:

- $S_g = \{s_1, s_2, \ldots, s_n\}$ = golden steps
- $S_i = \{r_1, r_2, \ldots, r_m\}$ = model steps

Define:

$$SLMS_i = \frac{\sum_{k=1}^{n} \max_j Sim(s_k, r_j)}{n}$$

Where:

- Sim = semantic similarity between steps

Measures:

Did the model logically cover the required reasoning steps?

Consistency Score (CS)

Run the same query T times:

Let:

$$CS_i = 1 - Var(RAS_i^1, RAS_i^2, ..., RAS_i^T)$$

Low variance = stable reasoning

High variance = unreliable reasoning

Low variance = stable reasoning

High variance = unreliable reasoning

Domain Knowledge Utilization Score (DKUS)

Let:

- C_g = required **domain concepts**
- C_i = **concepts** extracted from model **reasoning**

Then: $$DKUS_i = \frac{|C_g \cap C_i|}{|C_g|}$$

Final Composite Performance Score (FPS)

$$FPS = w_1AAS + w_2RAS + w_3SLMS + w_4CS + w_5DKUS$$

Where: $\sum w_k = 1$

Weights (w_k) are user-defined based on priority:

- accuracy-focused
- reasoning-focused
- domain-knowledge-focused

TechStack

LLM APIs
Integration with multiple Large Language Models for response generation

Backend Framework
Python-based evaluation pipeline

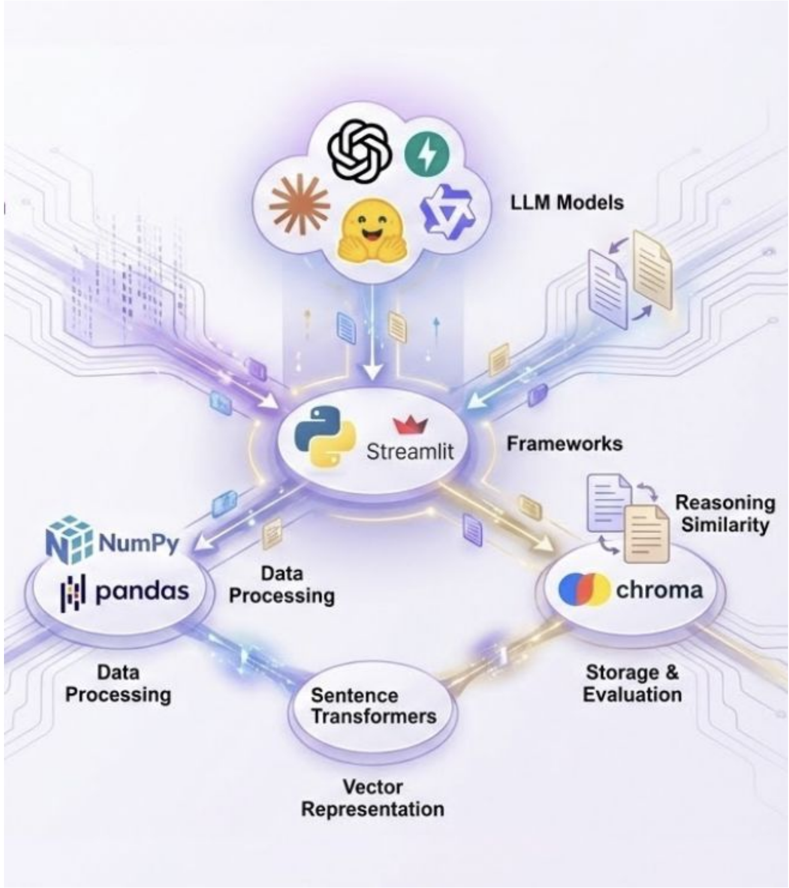
Frontend Framework
Streamlit Based

Data Processing
Pandas / NumPy for response handling and preprocessing

Reasoning Similarity Evaluation
Embedding-based semantic similarity techniques

Vector Representation
Sentence Transformers for encoding reasoning steps

Storage Layer
Chroma DB Database for storing prompts, responses, and evaluation results



Expected Results

1. overall ranking of LLMs by effectiveness
2. accuracy scores for final responses
3. performance evaluations for each LLM (for general and domain effectiveness)
4. guidance on selecting the best model for user applications
5. comparison of model reasoning with benchmark reasoning

Conclusion

This project proposes a structured approach to evaluating Large Language Models by comparing their generated reasoning and answers against a verified golden truth dataset. The evaluation framework developed through this project will enable meaningful comparison between models and support informed decision-making when selecting an LLM for practical use cases.

THANKYOU !